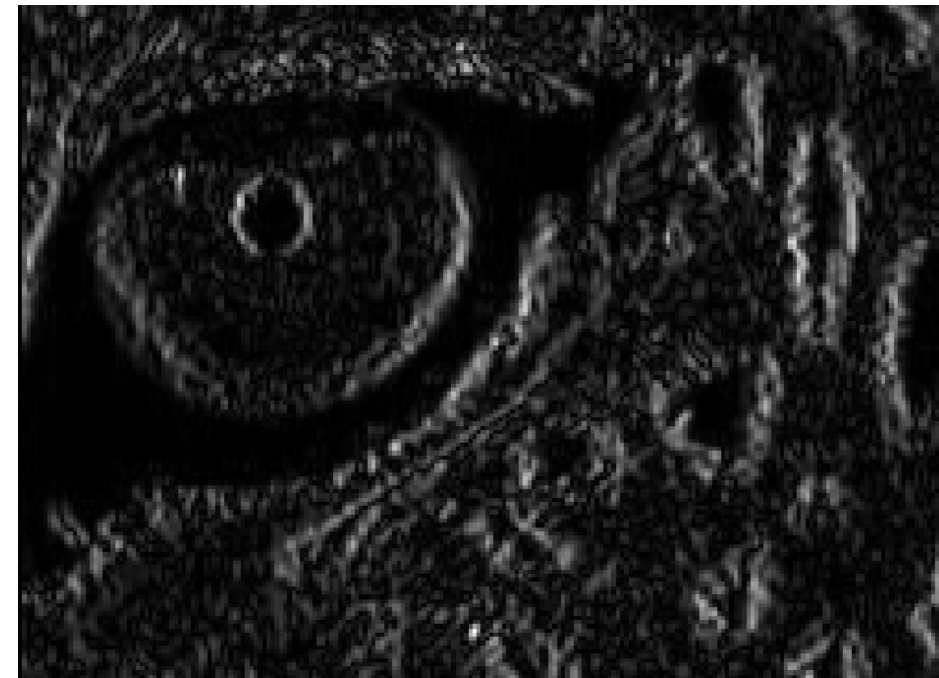


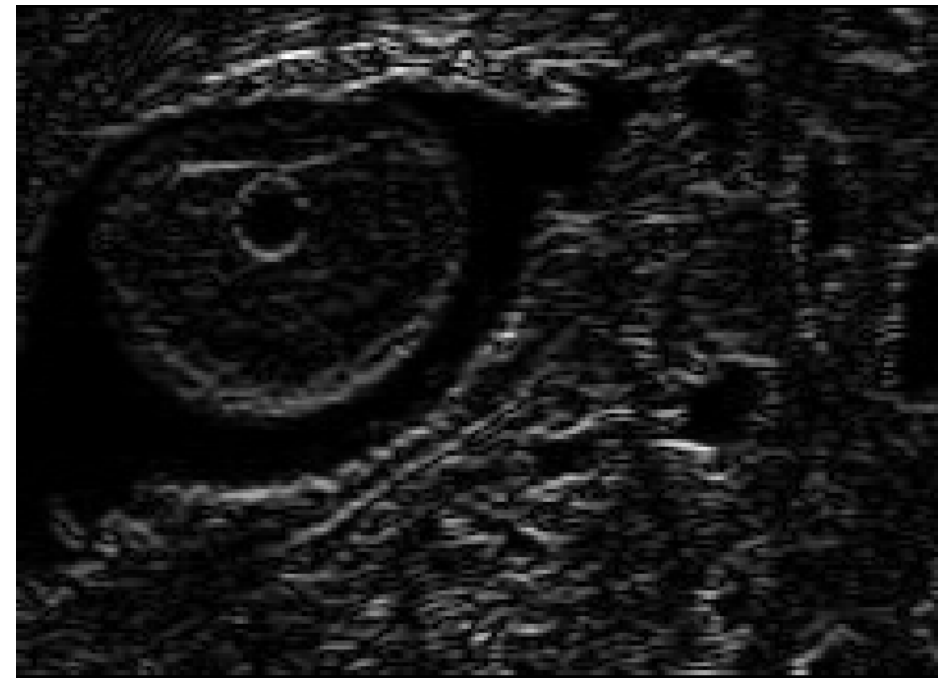
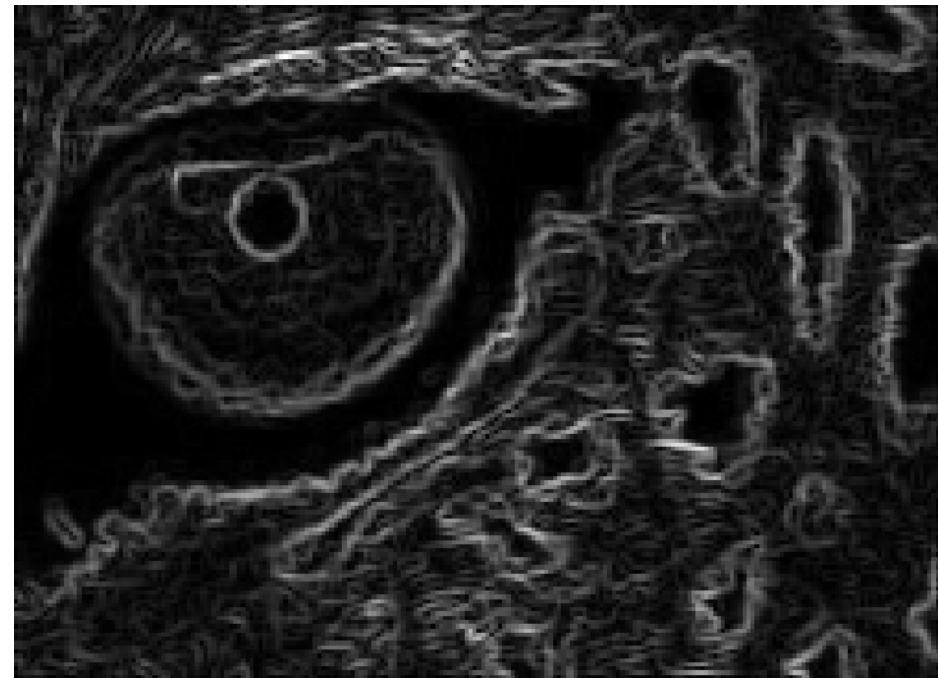
Original



Gx (vertical edges)



Gy (horizontal edges)

Magnitude $|\nabla f|$ Direction θ (mod 180°)